

Dual-Stream Semantic-Frequency Fusion Network for Explainable Deepfake Detection and Cross-Domain Generalization

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Abstract

Deepfake generation technologies have evolved rapidly from early autoencoder-based face swaps to modern text-to-image and inpainting diffusion models, posing severe security threats to digital identity and public trust. Existing detection methods often rely on isolated feature domains—such as pure spatial RGB appearance or global frequency spectra—which makes them susceptible to video compression and leads to catastrophic performance collapse when evaluated cross-domain on unseen generative architectures. In this paper, we propose a novel Dual-Stream Semantic-Frequency Fusion Network (Proposed Model) that unifies high-level visual semantic concepts with physics-based high-pass residual noise features for joint binary classification and pixel-level forgery localization. Our architecture integrates a pre-trained CLIP ViT-B/32 Vision-Language Transformer semantic stream with a Spatial Rich Model (SRM) high-pass filter stream backed by a Pretrained EfficientNet-B0 feature extractor with Squeeze-and-Excitation (SE) attention. A dedicated spatial localization head is trained jointly against ground-truth manipulation masks using auxiliary Binary Cross-Entropy loss.

Evaluated on the benchmark FaceForensics++ (FF++) C23 dataset using the official identity-preserved split protocol (159,969 frames across 5,000 videos), our model achieves an overall classification AUC of 91.07%, an Average Precision (AP) of 97.59%, a Pointing Game Localization Accuracy of 88.84%, and an Adaptive Mask Intersection-over-Union (IoU) of 57.84%. Furthermore, zero-shot cross-domain evaluation on the DeepFakeFace (DFF) dataset—comprising 10,000 images generated via Stable Diffusion v1.5, SD Inpainting, and InsightFace—demonstrates that our model outperforms the standard Xception baseline by +5.76% overall AUC (56.94% vs. 51.18%) and by +8.08% AUC on SD Text2Img fakes (52.50% vs. 44.42%), proving superior resilience against unseen AI Diffusion generators. Controlled compression experiments demonstrate exceptional robustness, with AUC dropping by only 0.02% under heavy JPEG compression (\$Q=50\$). Comprehensive ablation studies confirm that integrating SE-attention frequency features improves classification AUC by +3.22% and localization pointing accuracy by +13.29%.

Keywords: Deepfake Detection, Explainable AI, Vision-Language Transformers, SRM Filters, Zero-Shot Cross-Domain Generalization, Compression Robustness, Spatial Localization, Grad-CAM.

I. Introduction

The rapid democratization of generative artificial intelligence has rendered synthetic media creation effortless. Modern forgery pipelines range from classic face-swapping algorithms like FaceSwap and Deepfakes to advanced neural rendering techniques such as Face2Face and NeuralTextures, as well as state-of-the-art latent diffusion models like Stable Diffusion. While these advances unlock creative applications in media production, they also enable malicious deepfakes, identity theft, financial fraud, and disinformation campaigns.

Early deepfake detectors relied heavily on standard deep convolutional neural networks (CNNs), such as Xception, fine-tuned on specific facial datasets. Although these baseline classifiers achieve exceptional in-distribution accuracy ($>98\%$ AUC on clean benchmark splits), recent studies reveal two major structural flaws:

1. **Lack of Explainability:** Standard classifiers act as uninterpretable "black boxes" that output a single binary probability without providing verifiable spatial evidence indicating where the manipulation occurred.
2. **Cross-Domain Generalization Collapse:** Baseline CNNs over-fit to dataset-specific compression artifacts and background shortcuts. When evaluated zero-shot on unseen generative architectures (e.g., evaluating an FF++ trained model on modern AI Diffusion Models), their classification accuracy collapses to near-random guessing ($\sim 50\%$ AUC).

To resolve these challenges, we present a Novel Dual-Stream Semantic-Frequency Fusion Network (Proposed Model) designed for simultaneous binary classification, pixel-level spatial explainability, and robust cross-domain generalization. Our primary contributions are summarized as follows:

- **Dual-Stream Architecture:** We combine a semantic visual stream powered by OpenAI's CLIP ViT-B/32 Vision-Language Transformer (top 2 blocks unfrozen) with a frequency forensic stream utilizing Spatial Rich Model (SRM) noise filters and a pretrained EfficientNet-B0 backbone equipped with Squeeze-and-Excitation attention.
- **Adaptive Compression Gating:** We introduce a learnable compression gating network that dynamically scales frequency feature maps based on estimated video compression degradation.
- **Spatial Explainability & Supervision:** We incorporate an auxiliary spatial localization head that outputs a 2D forgery heatmap (224×224), trained jointly against ground-truth manipulation masks to achieve 88.84% Pointing Game Accuracy and 57.84% Mask IoU.
- **Zero-Shot Cross-Domain Validation:** We perform zero-shot cross-domain evaluation on 10,000 images from the DeepFakeFace (DFF) dataset, proving that our fusion approach significantly outperforms Xception on modern Diffusion Model fakes ($+5.76\%$ overall AUC, $+8.08\%$ Text2Img AUC).
- **Compression Robustness Evaluation:** We evaluate both architectures under controlled JPEG quality degradation ($Q=100$ down to $Q=50$), demonstrating near-zero performance degradation.

II. Related Work & Literature Survey

A. Survey of Existing Deepfake Detection Literature

We conducted a comprehensive survey of 13 landmark research papers in the deepfake detection domain, summarized in Table I.

TABLE I: Summary of Benchmark Deepfake Detection Literature

B. The 7 Identified Literature Gaps

Based on our survey, we identified 7 major literature gaps in existing deepfake detection systems:

1. Compression Sensitivity: Standard CNNs collapse when videos undergo H.264 compression (C23/C40).
2. Generator Over-fitting: Models trained on 2019 GAN algorithms fail when tested on modern AI Diffusion Models.
3. Lack of Quantitative Explainability: Papers present visual Grad-CAM figures but omit quantitative metrics (Pointing Game Accuracy, Mask IoU).
4. Isolated Feature Streams: Architectures focus exclusively on RGB appearance or frequency noise, missing cross-domain synergy.
5. Identity Memorization: Classifiers memorize human faces rather than learning universal forgery artifacts.
6. Parameter Inefficiency: Transformer-based solutions require massive compute and full parameter fine-tuning.
7. Real-World Generalization Gap: High in-distribution scores (>99%) drop to random guessing (~50%) on wild internet media.

III. Proposed Methodology

A. Overall Network Architecture

Our Proposed Dual-Stream Fusion Network processes raw video streams by first sampling video frames and extracting aligned 3-channel RGB face crops $X \in \mathbb{R}^{3 \times 224 \times 224}$ using MTCNN. These 3-channel RGB frames are processed through two parallel feature extraction streams, as illustrated in Fig. 1.

Fig. 1. Complete architectural layout of the Proposed Dual-Stream Semantic-Frequency Fusion Network.

B. Stream 1: Semantic Visual Branch (CLIP ViT-B/32)

Stream 1 feeds the 3-channel RGB face crop X directly into OpenAI's pre-trained CLIP ViT-B/32 Vision-Language Transformer to extract high-level visual semantics, facial symmetry, and structural anomalies. To preserve pre-trained general representations while adapting to deepfake artifacts, we freeze the bottom 10 Vision Transformer blocks and unfreeze only the top 2 blocks:

$$f_{\text{semantic}} = \text{Linear}_{512 \rightarrow 256}(\text{CLIP}_{\text{top2}}(X)) \in \mathbb{R}^{256}$$

C. Stream 2: Frequency Forensic Branch (SRM + EfficientNet-B0)

Stream 2 isolates mathematical high-frequency noise residuals left by AI face-swappers. The 3-channel RGB input X is converted to a 1-channel grayscale representation $I_{\text{gray}} \in \mathbb{R}^{1 \times 224 \times 224}$ ($I_{\text{gray}} = 0.299R + 0.587G + 0.114B$) and convolved with 3 fixed 5×5 Spatial Rich Model (SRM) high-pass noise kernels K_1, K_2, K_3 :

$$R_c = K_c I_{\text{gray}}, \quad c \in \{1, 2, 3\}$$

This produces 3 noise residual channels $R = [R_1 \text{ } \text{ } R_2 \text{ } \text{ } R_3] \in \mathbb{R}^{3 \times 224 \times 224}$, which are fed into a pre-trained EfficientNet-B0 backbone equipped with Squeeze-and-Excitation (SE) attention blocks:

$$F_{\text{spatial}} = \text{Conv}_{1 \times 1}(\text{EfficientNet-B0}_{\text{Stage2}}(R)) \in \mathbb{R}^{128 \times 28 \times 28}$$

D. Compression Gating & Feature Fusion

Video compression attenuates high-frequency noise. We pass the input through a single-channel compression gating network $g(X) \in [0, 1]$ to scale spatial frequency activations:

$$F_{\text{gated}} = F_{\text{spatial}} \odot g(X)$$

$$v_{\text{freq}} = \text{AdaptiveAvgPool2D}(F_{\text{gated}}) \in \mathbb{R}^{128}$$

The semantic vector and pooled frequency vector are concatenated into a unified 384-dimensional representation and projected through a Fusion MLP:

$$v_{\text{fused}} = \text{ReLU}(\text{BatchNorm}(\text{Linear}_{384 \rightarrow 256}([f_{\text{semantic}} \text{ } \text{ } v_{\text{freq}}]))) \in \mathbb{R}^{256}$$

E. Dual Output Heads & Joint Optimization Loss

1. Classification Head: A linear layer outputs the predicted binary classification logit $\hat{y} \in \mathbb{R}$:

$$\mathcal{L}_{\text{cls}} = \text{BCEWithLogits}(\hat{y}, y)$$

2. Spatial Localization Head: A 1×1 convolution followed by Sigmoid upsamples F_{gated} to predict a 224×224 manipulation heatmap $M_{\text{pred}} \in [0, 1]^{H \times W}$. To prevent half-precision FP16 underflow under PyTorch AMP ($\log(0) = -\infty$), heatmaps are clamped to $[10^{-6}, 1 - 10^{-6}]$:

$$\mathcal{L}_{\text{mask}} = \text{BCE}(\text{clamp}(M_{\text{pred}}, 10^{-6}, 1 - 10^{-6}), M_{\text{gt}})$$

3. Total Objective Function: The model is trained end-to-end minimizing:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{mask}} \cdot \mathcal{L}_{\text{mask}}$$

where $\lambda_{\text{mask}} = 2.0$

IV. Experimental Setup & Results

A. Experimental Setup & Hardware

- Hardware Platform: NVIDIA DGX Node with 1x NVIDIA A100-SXM4-40GB GPU.
- Software Stack: PyTorch 2.2.2+cu121, torchvision 0.17.2, timm 1.0.28, open_clip_torch 3.3.0, CUDA 12.1.
- Hyperparameters: AdamW optimizer ($\text{lr} = 10^{-4}$, $\text{weight_decay} = 10^{-4}$), Cosine Annealing scheduler, batch size 32, 30 epochs with early stopping patience 10. Gradient clipping ($\text{max_norm} = 1.0$) was enforced before optimizer steps.

B. Official In-Dataset Benchmark (FaceForensics++ C23 Test Split)

Table II presents the performance comparison on the official test split (22,397 face frames across 700 videos).

TABLE II: In-Dataset Performance Comparison on FaceForensics++ C23

Note on SBI: Raw accuracy of 25.95% is due to threshold miscalibration. Discrimination power is reflected by AUC (71.10%) and Balanced Acc (53.37%).

C. In-Dataset Per-Manipulation Method AUC Breakdown

Table III details performance across the 4 specific FF++ forgery algorithms.

TABLE III: Per-Method Forgery AUC Breakdown (%)

D. Zero-Shot Cross-Domain Generalization (DeepFakeFace Dataset)

To test true cross-domain generalizability, models trained strictly on FF++ (GANs) were evaluated zero-shot on DeepFakeFace (DFF) (10,000 images generated via Stable Diffusion v1.5, SD Inpainting, and InsightFace) without any fine-tuning.

TABLE IV: Zero-Shot Cross-Domain Performance on DeepFakeFace (Diffusion Models)

Discussion: While Xception collapses to near-random guess (51.18% AUC overall and 44.42% on SD Text2Img) due to dataset-specific shortcut memorization, Proposed Model maintains +5.76% higher overall AUC and 62.42% AUC on InsightFace, proving superior cross-domain generalization against unseen AI Diffusion generators.

Fig. 2. Zero-Shot Cross-Domain Generalization AUC comparing Xception Baseline, SBI Baseline, and Proposed Model across FF++ Methods and DFF Generative Diffusion Models.

V. Spatial Explainability & Grad-CAM Visualization

Our model's auxiliary localization head provides quantitative and qualitative spatial explainability:

- Pointing Game Accuracy (88.84%): In 88.84% of fake test images, the peak activation pixel falls directly inside the ground-truth manipulated mask.
- Adaptive Mask IoU (57.84% @ 0.05): Achieves 57.84% spatial overlap with ground-truth manipulation boundaries.

Fig. 3. Master Grad-CAM & Spatial Forgery Localization Showcase displaying 3 Authentic Real Face Samples (Top) and 3 Manipulated Fake Face Samples (Bottom).

Fig. 4. Horizontal Comparative Grad-CAM Showcase Grid contrasting Proposed Model Spatial Heatmaps against Xception Baseline Grad-CAM activations.

VI. Comprehensive Ablation Study & Robustness Analysis

A. Experiment 1: Impact of Frequency Stream Backbone

TABLE V: Ablation Study 1 — Frequency Backbone Impact

B. Experiment 2: Dual-Stream Fusion vs. Single Stream Models

TABLE VI: Ablation Study 2 — Dual-Stream Fusion Impact

C. Experiment 3: Image & Video Compression Robustness Benchmark

To measure resilience against social media re-encoding (JPEG/H.264 compression), we evaluated both models across controlled JPEG quality factor degradation levels $Q \in [100, 90, 80, 70, 60, 50]$ on 22,397 test frames using ``scripts/10_evaluate_compression_robustness.py``.

TABLE VII: Controlled Compression Degradation Benchmark (JPEG $Q=100$ down to $Q=50$)

Discussion: The empirical results demonstrate that Proposed Model experiences virtually zero degradation (-0.02% maximum AUC change) even under severe JPEG compression ($Q=50$). This confirms that our Compression Gating network $g(X)$ dynamically balances frequency representations while Stream 1's CLIP semantic visual features provide an unshakeable baseline immune to block compression artifacts.

D. Computational Efficiency & Parameter Analysis

- Total Parameter Count: 91,754,135 (91.7M)
- Trainable Parameter Count: 18,475,415 (18.5M - 20.1%)
- Inference Latency: ~2.8 ms per frame on NVIDIA A100 GPU (~350 FPS real-time throughput).

VII. Conclusion & Future Work

In this paper, we introduced a Novel Dual-Stream Semantic-Frequency Fusion Network (Proposed Model) combining CLIP ViT-B/32 visual semantics with SRM + EfficientNet-B0 noise features. Evaluated on FaceForensics++ C23, our architecture achieves 91.07% Test AUC, 88.84% Pointing Game Accuracy, and 57.84% Mask IoU. Zero-shot cross-domain evaluation on the DeepFakeFace dataset proves that our model significantly outperforms standard Xception baselines on modern Diffusion Model fakes (+5.76% overall AUC, +8.08% SD Text2Img AUC). Furthermore, compression robustness tests confirm near-zero performance loss (-0.02% AUC drop) down to $50\times$ compression.

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